

Feasibility Study & Proposal

Project Title:

Dynamic Optimization of RO Plant Energy Consumption using Machine Learning

Objective:

To develop a machine learning-based soft sensor that dynamically minimizes energy consumption (E_c) while maintaining target water quality and water production levels by adapting to influent and membrane conditions over time.

Problem Context & Business Case:

- Current fixed set-points lead to suboptimal energy use.
- Human error and lack of tuning time lead to outages and inefficiencies.
- A dynamic ML model will provide decision support and reduce unplanned downtime.

Dataset Overview:

Features include:

- Operational: Q_{in} , Q_{out} , E_c
- Water Quality: Turbidity (Feed/Product), TOC (Feed/Product), EC, NH_4 , BOD, COD, TN
- Temporal: Year, Month, Day, Temperature

Feasibility Highlights:

1. Energy Consumption Prediction
2. Feature Importance Insights using RF & XGBoost
3. Operational Guidance via scenario classification

Proposed Methodology:

1. EDA: Visual patterns, outlier detection
2. Feature Engineering: Lag features, deltas
3. Feature Selection: RF, XGBoost
4. Modeling: KNN, XGBoost, LightGBM, GPR, SVR
5. Evaluation: RMSE, MAE, MAPE, J_2
6. Insights: Feature trends over time

Key Deliverables:

- Dynamic energy prediction models
- Feature importance & impact reports
- Temporal analytics (seasonal impacts)
- Recommendations for dynamic tuning
- Optional dashboard integration

Tools & Stack:

Feasibility Study & Proposal

- Python (pandas, scikit-learn, xgboost, lightgbm)
- Jupyter or Streamlit for visualization
- Plotly/Matplotlib for charts
- Optional: Flask for API integration

Why This Approach?

- Builds on proven research (e.g., Melbourne WWTP)
- Fast models like KNN/RF suitable for real-time ops
- Tight focus on actionable insights in a 2-day window